Q3a) What is Association Rule Mining? Explain Apriori Algorithm with Example.

Association Rule Mining

Association Rule Mining is a data mining technique used to discover relationships or associations between items in large datasets.

It is commonly used in: Market Basket Analysis, Recommendation Systems, Customer Purchase Analysis

Basic Terms

1. Support - Support shows how frequently an itemset appears in the dataset. $Support(A \rightarrow B) = \frac{Transactions\ containing\ A\ and\ B}{Total\ Transactions}$

2. Confidence - Confidence shows how often items in B appear in transactions containing A. $Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)}$

Apriori Algorithm

Apriori is a popular algorithm used to find frequent itemsets and generate association rules. It works on the principle: "If an itemset is frequent, all its subsets must also be frequent."

Steps of Apriori Algorithm

Step 1: Generate Frequent 1-itemsets

Find items whose support is greater than minimum support.

Step 2: Generate Candidate Itemsets

Combine frequent itemsets to generate larger itemsets.

Step 3: Prune Infrequent Itemsets

Remove itemsets whose support is below threshold.

Step 4: Repeat

Continue until no more frequent itemsets are found.

Example of Apriori Algorithm -

Transaction Dataset :-

Transaction ID	**Items**
T1	Milk, Bread, Butter
T2	Bread, Butter
T3	Milk, Bread
T4	Milk, Butter
T5	Bread, Butter

Assume minimum support = 2

Step 1: Count Individual Items

Item	Support Count
Milk	3
Bread	4
Butter	4

All are frequent.

Step 2: Generate 2-itemsets

Itemset	Support Count
Milk, Bread	2
Milk, Butter	2
Bread, Butter	3

All are frequent.

Step 3: Generate 3-itemset

Itemset	Support Count
Milk, Bread, Butter	1

Not frequent because support < 2.

Association Rule Example : Bread → Butter

Customers buying bread are likely to buy butter.

Applications of Association Rule Mining - Product recommendation, Market basket analysis, Medical diagnosis, Website usage analysis

Association Rule Mining identifies relationships between items, and the Apriori algorithm efficiently finds frequent itemsets and useful association rules.

Q3b) Decision Tree Construction using Entropy and Information Gain

Decision Tree - A Decision Tree is a supervised machine learning algorithm used for classification and prediction.

It divides data into branches based on attribute selection.

Entropy - Entropy measures impurity or randomness in data.

$$Entropy(S) = -p_1 \log_2 p_1 - p_2 \log_2 p_2$$

- Entropy = 0 → Pure data
- Higher entropy → More disorder

Information Gain - Information Gain measures reduction in entropy after splitting data.

$$Information\ Gain = Entropy(Parent) - Weighted\ Entropy(Children)$$

Attribute with highest Information Gain is selected as root node.

Steps in Decision Tree Construction -

1. Calculate entropy of dataset.
2. Compute information gain for each attribute.
3. Select attribute with highest gain.
4. Split dataset.
5. Repeat recursively until pure classification is achieved.

Small Example - Dataset

Outlook	**Play Cricket**
Sunny	No
Rainy	Yes
Sunny	No
Overcast	Yes

Step 1: Calculate Entropy

There are: Yes = 2 & No = 2

Entropy is maximum because dataset is mixed.

Step 2: Choose Attribute

Suppose "Outlook" gives highest information gain.

Step 3: Create Decision Tree

```
        Outlook
       /   |    \
  Sunny Rainy Overcast
   No    Yes    Yes
```

Decision Trees use entropy and information gain to select the best attribute for splitting data and creating accurate classification models.

Q4a) Explain Naïve Bayes Classifier and its Applications

Naïve Bayes Classifier - Naïve Bayes is a supervised machine learning classification algorithm based on Bayes' Theorem.

It assumes that all features are independent of each other.

Bayes Theorem : $P(A \mid B) = \frac{P(B|A)P(A)}{P(B)}$

Where: $P(A \mid B)$= Posterior Probability, $P(B \mid A)$= Likelihood, $P(A)$= Prior Probability, $P(B)$= Evidence

Working of Naïve Bayes

1. Calculate prior probabilities.
2. Calculate likelihood probabilities.
3. Apply Bayes theorem.
4. Predict class with highest probability.

Applications - Spam email filtering, Sentiment analysis, Medical diagnosis, Document classification, Recommendation systems

Advantages - Simple and fast, Works well with large datasets, Good for text classification

Naïve Bayes is an efficient probabilistic classifier widely used in text mining and prediction tasks.

Q4b) Logistic Regression for Loan Default Prediction

Logistic Regression is a supervised learning algorithm used for binary classification problems.

It predicts probability values between 0 and 1.

Logistic Function $P(Y = 1) = \frac{1}{1+e^{-(b_0+b_1x_1+b_2x_2)}}$

Where: $Y = 1$→ Loan Default, x_1→ Loan Amount, x_2→ Default History

Dataset Example -

Loan Amount	**Default History**	**Loan Default**
High	Yes	Yes
Low	No	No

Loan Amount	Default History	Loan Default
Medium	Yes	Yes
Low	No	No

Working

Step 1: Input Features

- Loan Amount
- Default History

Step 2: Train Logistic Regression Model

Model learns relationship between inputs and output.

Step 3: Predict Probability

For a new customer:

Loan Amount	Default History
High	Yes

Model predicts probability of default.

If probability > 0.5 → Default = Yes

Otherwise → No

Applications of Logistic Regression - Loan approval systems, Fraud detection, Disease prediction, Spam filtering

Logistic Regression is widely used for binary classification problems such as loan default prediction because it predicts probabilities effectively and is easy to interpret.

Q3a) Define and Explain Entropy and Information Gain. Calculate the Entropy of the Given Distribution.

Entropy - Entropy is a measure of impurity, randomness, or uncertainty in a dataset.

- Higher entropy → More disorder
- Lower entropy → More pure data

Entropy is mainly used in Decision Tree algorithms.

Entropy Formula : $Entropy(S) = -\sum p_i \log_2 p_i$

Where: p_i= Probability of each class

Information Gain - Information Gain measures the reduction in entropy after splitting the dataset using an attribute.

It helps in selecting the best attribute in Decision Tree construction.

Information Gain Formula

$$Information\ Gain = Entropy(Parent) - Weighted\ Entropy(Children)$$

- Higher Information Gain → Better attribute for splitting.

Given Distribution - ...

Step 1: Find Total Counts

Sweet Fruits: 10 + 5 = 15

Sour Fruits: 15 + 5 = 20

Total Fruits: 15 + 20 = 35

Step 2: Calculate Probabilities

Probability of Sweet: $P(Sweet) = \frac{15}{35} = 0.4286$

Probability of Sour: $P(Sour) = \frac{20}{35} = 0.5714$

Step 3: Calculate Entropy: $Entropy(S) = -0.4286\log_2(0.4286) - 0.5714\log_2(0.5714)$

Step 4: Final Calculation: Entropy(S) ≈ 0.985

Final Answer: Entropy of Given Distribution - Entropy ≈ 0.985

This indicates the dataset is highly mixed (impure).

Entropy measures uncertainty in data, while Information Gain measures reduction in uncertainty after splitting. These concepts are widely used in Decision Tree algorithms for classification tasks.

Q4b) Describe Different Categories of Analytics

Analytics is classified into different categories based on the type of insights generated from data.

Categories of Analytics -

1. Descriptive Analytics

Purpose: Describes past events and summarizes historical data.

Question Answered: "What happened?"

Examples: Sales reports, Attendance reports, Website traffic analysis

Techniques Used: Reporting, Data visualization, Dashboards

2. Diagnostic Analytics

Purpose: Analyzes data to find causes of problems or events.

Question Answered: "Why did it happen?"

Examples: Finding reasons for sales decline, Identifying causes of machine failure

Techniques Used: Drill-down analysis, Correlation analysis, Data mining

3. Predictive Analytics

Purpose: Predicts future outcomes using historical data.

Question Answered: "What may happen in future?"

Examples: Weather forecasting, Stock market prediction, Sales forecasting

Techniques Used: Machine learning, Regression, Decision trees

4. Prescriptive Analytics

Purpose: Suggests actions and solutions for decision-making.

Question Answered: "What should be done?"

Examples: Route optimization, Business strategy recommendations, Automated decision systems

Techniques Used: Optimization algorithms, AI systems, Simulation models

Diagram of Analytics Categories -

Descriptive → Diagnostic → Predictive → Prescriptive
↓ ↓ ↓ ↓
What happened? Why happened? What will happen? What should be done?

The four categories of analytics—Descriptive, Diagnostic, Predictive, and Prescriptive—help organizations understand past events, identify causes, predict future trends, and make intelligent decisions.

Q3a) What is Logistic Regression? How does it differ from Linear Regression? What is Sigmoid Function and its role in Logistic Regression?

Logistic Regression is a supervised machine learning algorithm used for classification problems. It predicts the probability of occurrence of an event such as:Spam or Not Spam, Pass or Fail, Fraud or Not Fraud.

The output of logistic regression lies between 0 and 1.

Logistic Regression Equation - $P(Y=1)=\frac{1}{1+e^{-(b_0+b_1x_1+b_2x_2+\cdots+b_nx_n)}}$

Where: $P(Y=1)$= Probability of positive class, b_0= Intercept, b_1, b_2= Coefficients, x_1, x_2= Features

Logistic Regression	**Linear Regression**
Used for classification	Used for prediction/regression
Output is categorical	Output is continuous
Output ranges from 0 to 1	Output can be any value
Uses sigmoid function	Uses straight-line equation
Predicts probabilities	Predicts numerical values
Ex - Predict whether an email is spam or not.	**Ex** - Predict house price based on area.

Sigmoid function converts any real value into a probability value between 0 and 1.

Sigmoid Function Formula: $\sigma(z)=\frac{1}{1+e^{-z}}$

Where: $z=b_0+b_1x_1+b_2x_2+\cdots+b_nx_n$

Role of Sigmoid Function in Logistic Regression

- Converts linear output into probability
- Helps in binary classification
- Produces S-shaped curve
- Maps values between 0 and 1

Sigmoid Curve - Input Value → Probability Output
Large Negative → Near 0
0 → 0.5
Large Positive → Near 1

Applications of Logistic Regression - Spam detection, Disease prediction, Loan approval, Fraud detection

Logistic Regression is widely used for binary classification problems. The sigmoid function plays a vital role by converting linear outputs into probability values between 0 and 1.

b) Naïve Bayes Classification for Spam Detection

Given Training Dataset ...

We need to classify:

Offer = 1
Free = 1

Step 1: Calculate Prior Probabilities

Total Emails: 5

Spam = Yes

Emails 2, 3, 5

P(Spam=Yes)=3/5=0.6

Spam = No

Emails 1, 4

P(Spam=No)=2/5=0.4

Step 2: Conditional Probabilities

For Spam = Yes

Among spam emails (2,3,5):

Offer = 1

Emails 3 and 5

P(Offer=1|Spam=Yes)=2/3

Free = 1

Emails 2,3,5

P(Free=1|Spam=Yes)=3/3=1

For Spam = No

Among non-spam emails (1,4):

Offer = 1

Email 1 only

P(Offer=1|Spam=No)=1/2

Free = 1

Email 4 only

P(Free=1|Spam=No)=1/2

Step 3: Apply Naïve Bayes Formula

Probability for Spam = Yes

$$P(Yes \mid Offer = 1, Free = 1) = P(Yes) \times P(Offer = 1 \mid Yes) \times P(Free = 1 \mid Yes)$$

Calculation: = 0.6 × (2/3) × 1 = 0.4

Probability for Spam = No

$$P(No \mid Offer = 1, Free = 1) = P(No) \times P(Offer = 1 \mid No) \times P(Free = 1 \mid No)$$

Calculation: = 0.4 × (1/2) × (1/2) = 0.1

Step 4: Final Classification

Since: 0.4 > 0.1

The new email is classified as: Final Answer Spam = Yes

Using Naïve Bayes classification, the probability of the new email being spam is higher for the "Yes" class. Therefore, the email is classified as Spam.

Q3b) Calculation of Support and Confidence for Possible Itemsets

Given Transactions ...

Step 1: Find Support Count of Individual Items-

Item	**SupportCount**
Onion	3
Potato	3
Cold Drink	4
Burger	2
Eggs	3
Milk	2

Step 2: Calculate Support Value

Support Formula:

$$Support(A) = \frac{Number\ of\ transactions\ containing\ A}{Total\ Transactions}$$

Total Transactions = 5

Support of 1-Itemsets

Item	**Support**
Onion	3/5 = 0.60
Potato	3/5 = 0.60
Cold Drink	4/5 = 0.80
Burger	2/5 = 0.40
Eggs	3/5 = 0.60
Milk	2/5 = 0.40

Step 3: Generate 2-Itemsets

Itemset	**Support Count**	**Support**
Onion, Potato	1	0.20
Onion, Cold Drink	3	0.60
Potato, Cold Drink	2	0.40
Burger, Cold Drink	2	0.40
Potato, Eggs	2	0.40
Potato, Milk	2	0.40

Itemset	Support Count	Support
Milk, Eggs	2	0.40
Eggs, Cold Drink	1	0.20
Burger, Potato	1	0.20
Onion, Eggs	1	0.20

Step 4: Calculate Confidence

Confidence Formula: $Confidence(A \rightarrow B) = \frac{Support(A \cup B)}{Support(A)}$

Confidence Calculations

1. **Onion → Cold Drink**

Confidence = Support(Onion, Cold Drink) / Support(Onion) = 3/3 = 1.0 = 100%

2. **Cold Drink → Onion** = 3/4 = 0.75 = 75%
3. **Potato → Milk** = 2/3 = 0.667 = 66.7%
4. **Milk → Potato** = 2/2 = 1.0 = 100%
5. **Potato → Eggs** = 2/3 = 66.7%
6. **Eggs → Potato** = 2/3 = 66.7%
7. **Burger → Cold Drink** = 2/2 = 100%
8. **Cold Drink → Burger** = 2/4 = 50%

Strong Association Rules

Rule	Confidence
Onion → Cold Drink	100%
Milk → Potato	100%
Burger → Cold Drink	100%

These are considered strong rules.

Q4a) Explain the Need of Logistic Regression along with its Various Types

Logistic Regression is a supervised machine learning algorithm used for classification problems where the output is categorical.

It predicts the probability of occurrence of an event such as: Yes/No, Spam/Not Spam, Pass/Fail. The output value lies between 0 and 1.

Need of Logistic Regression -

1. Binary Classification Problems - Used when output has two classes.

Examples - Loan Approved or Rejected, Disease Present or Absent

2. Probability Prediction - Predicts probability instead of direct values.

Example - Probability of customer buying a product.

3. Fast and Simple Model - Easy to implement, Computationally efficient

4. Interpretable Results - Relationship between features and output can be understood easily.

5. Works Well with Large Datasets - Efficient for real-world classification tasks.

Logistic Regression Equation - $P(Y = 1) = \frac{1}{1+e^{-z}}$

Where: $z = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$

Types of Logistic Regression -

1. Binary Logistic Regression

Used when output has only two classes.

Examples - Spam / Not Spam, Pass / Fail

2. Multinomial Logistic Regression

Used when output has more than two unordered classes.

Examples - Car Type: SUV, Sedan, Hatchback, Product Category Prediction

3. Ordinal Logistic Regression

Used when output categories have a meaningful order.

Examples - Ratings: Poor, Good, Excellent, Education Level

Applications of Logistic Regression - Spam email detection, Fraud detection, Disease prediction, Customer churn prediction, Loan approval systems

Logistic Regression is widely used for classification problems because it predicts probabilities effectively and provides simple, interpretable, and accurate results.

Q4b) Explain the Following Terms with Suitable Example

i) Removing Duplicates from Dataset

Duplicate data means repeated records present multiple times in a dataset.

Removing duplicates is the process of identifying and deleting repeated records to improve data quality.

Need for Removing Duplicates - Improves data accuracy, Reduces storage space, Prevents incorrect analysis, Improves model performance

Example –

Dataset Before Removing Duplicates

Student ID	Name	Marks
101	Amit	80
102	Ravi	75
101	Amit	80

Here, record of Amit is repeated.

Dataset After Removing Duplicates

Student ID	Name	Marks
101	Amit	80
102	Ravi	75

Duplicate record is removed.

Methods for Removing Duplicates - SQL DISTINCT keyword, Data cleaning tools, Python pandas drop_duplicates()

Removing duplicates improves data consistency and ensures accurate analytics results.

ii) Handling Missing Data

Missing data refers to unavailable or empty values in a dataset.

Handling missing data is important because missing values may reduce accuracy of analysis and machine learning models.

Causes of Missing Data - Data entry errors, Sensor failure, Incomplete forms, Data corruption

Methods for Handling Missing Data

1. Remove Missing Records - Delete rows containing missing values.

Suitable when: Very few records are missing.

2. Replace with Mean/Median/Mode - Missing values are replaced using statistical values.

Example - Replace missing age with average age.

3. Use Default Value - Replace missing value with constant value like: 0 or Unknown

4. Predict Missing Values - Use machine learning algorithms to estimate missing values.

Example –

Dataset Before Handling Missing Data

Employee	Salary
A	30000
B	NULL
C	25000

After Replacing Missing Value

Average Salary:

(30000 + 25000)/2 = 27500

Employee	Salary
A	30000
B	27500
C	25000

Handling missing data is an essential data preprocessing step that improves data quality, reliability, and analytical accuracy.

Q3a) Explain Why Decision Trees are Used. Draw a Sample Decision Tree and Explain its Parts.

A Decision Tree is a supervised machine learning algorithm used for: Classification, Prediction, Decision making. It represents decisions in the form of a tree-like structure.

Why Decision Trees are Used -

1. Easy to Understand - Decision trees are simple and visually understandable.

2. Used for Classification and Prediction - Can classify data into categories and predict outcomes.

3. Handles Large Datasets - Works efficiently with large amounts of data.

4. Requires Less Data Preparation - No need for normalization or scaling.

5. Supports Decision Making - Provides clear decision rules.

6. Handles Numerical and Categorical Data - Can work with different data types.

Applications of Decision Trees - Medical diagnosis, Loan approval, Fraud detection, Customer classification, Weather prediction

Sample Decision Tree

```
      Weather
    /  |       \
 Sunny  Rainy  Cloudy
  |      |          |
 Humidity  Wind    Play
 /      \       |
High  Normal Strong
 No     Yes    No
```

Parts of Decision Tree

1. Root Node

- Starting point of tree.
- Represents the main attribute.

Example: Weather

2. Decision Node - Intermediate node where decision is made.

Example: Humidity

3. Branch - Represents outcome of decision.

Example: Sunny → Humidity

4. Leaf Node - Final output or class label.

Example: Yes / No

Advantages of Decision Trees - Easy visualization, Fast decision making, Good accuracy, Easy interpretation

Decision Trees are widely used because they are simple, effective, and easy to interpret for classification and prediction tasks.

Q4a) What is Data Preprocessing? Explain Handling Missing Data and Data Transformation.

Data Preprocessing is the process of cleaning and preparing raw data before analysis or model building.

Raw data may contain: Missing values, Duplicate data, Noise, Inconsistent formats, Preprocessing improves data quality and model accuracy.

Steps in Data Preprocessing

Data Cleaning
↓
Handling Missing Data
↓
Data Integration
↓
Data Transformation
↓
Data Reduction

1. Handling Missing Data ...

2. Data Transformation - Data Transformation is the process of converting data into a suitable format for analysis and machine learning.

Types of Data Transformation -

i) Normalization - Scales data into a fixed range such as: 0 to 1

ii) *Aggregation* - Combining multiple data values into summary form.

Example: Monthly sales summary.

iii) *Generalization* - Low-level data is converted into higher-level concepts.

Example: Age → Young, Adult, Senior

iv) Encoding - Convert categorical data into numerical form.

Example:

Gender	**Encoded Value**
Male	1
Female	0

v) Smoothing - Removes noise from data.

Importance of Data Transformation - Improves model performance, Makes data suitable for algorithms, Reduces inconsistencies

Data preprocessing is an important step in Data Analytics that improves data quality. Handling missing data and transforming data properly helps in generating accurate and efficient analytical models.

Q3a) Explain Various Data Preprocessing Steps. Discuss Essential Python Libraries for Preprocessing.

Essential Python Libraries for Preprocessing

1. NumPy - Provides numerical operations and array handling.

Features - Mathematical functions, Multi-dimensional arrays

Example - import numpy as np

2. Pandas - Used for data manipulation and preprocessing.

Features – DataFrames, Handling missing data, Data filtering

Example - import pandas as pd

3. Scikit-learn - Provides preprocessing and machine learning tools.

Features – Normalization, Label encoding, Feature scaling

Example - from sklearn.preprocessing import StandardScaler

4. Matplotlib - Data visualization library.

Features – Graphs, Charts, Histograms

Example - import matplotlib.pyplot as plt

5. Seaborn - Advanced statistical visualization.

6. SciPy - Scientific and statistical computations.

Data preprocessing improves data quality and analytical accuracy. Python libraries such as NumPy, Pandas, Scikit-learn, and Matplotlib provide powerful tools for efficient preprocessing and analysis.

b) Explain Scikit-learn Library and Matplotlib with Example

1. Scikit-learn Library

Scikit-learn is an open-source Python machine learning library used for: Classification, Regression, Clustering, Data preprocessing

It provides simple and efficient tools for machine learning and analytics.

Features of Scikit-learn - Easy to use, Supports preprocessing, Provides ML algorithms, Model evaluation tools

Applications - Spam detection, Recommendation systems, Prediction models, Data classification

Example of Scikit-learn - Standardization Example

```
from sklearn.preprocessing import StandardScaler
import numpy as np

data = [[10], [20], [30]]

scaler = StandardScaler()
result = scaler.fit_transform(data)

print(result)
```

Output

Data gets normalized around mean 0.

2. Matplotlib Library

Matplotlib is a Python library used for data visualization and plotting graphs.

It helps represent data visually.

Features of Matplotlib - Line charts, Bar graphs, Pie charts, Histograms, Scatter plots

Applications - Data visualization, Statistical analysis, Reporting

Example of Matplotlib -

```
import matplotlib.pyplot as plt

x = [1,2,3,4]
y = [10,20,25,30]

plt.plot(x,y)
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.title("Sample Graph")

plt.show()
```

Output

A line graph is displayed showing relationship between X and Y values.

Difference Between Scikit-learn and Matplotlib

Scikit-learn	Matplotlib
Used for machine learning	Used for visualization
Provides ML algorithms	Provides plotting tools
Supports preprocessing	Supports charts and graphs

Scikit-learn is widely used for machine learning and preprocessing, while Matplotlib is used for visualizing data through graphs and charts. Together they play an important role in Data Science and Analytics.